

Introduction to Graph Theory

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1 Basic stuff about graphs

Definition 1.1 (Undirected graph): An (undirected) *graph* is a pair $G = (V, E)$ where

- V is the set of *vertices*;
- E is the set of *edges*, i.e. sets $\{v_1, v_2\}$ where v_1 and v_2 are vertices.

Example: Figure 1 shows a graph $G = (V, E)$ with

- $V = \{a, b, c, d, e, f, g\}$,
- $E = \{\{a, b\}, \{a, c\}, \{b, c\}, \{c, d\}, \{c, e\}, \{c, h\}, \{e, f\}, \{e, g\}, \{e, h\}, \{f, g\}\}$.

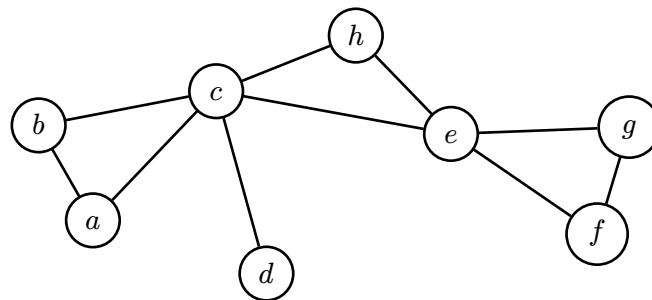


Figure 1: Example of a random graph.